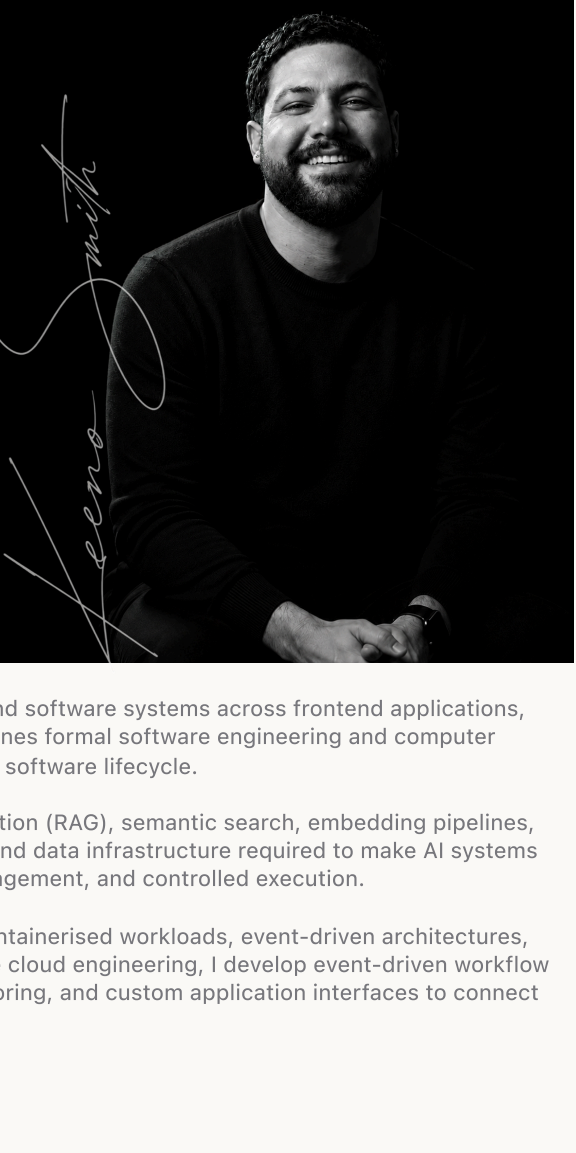


github.com/keenosmith-del
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Keeno Smith

Full-Stack Software Engineer & AI Engineer

AI Application Development · Generative & Agentic AI · Cloud Engineering · DevOps & Automation

Full-Stack Software Engineer & Web Developer specialising in the design, development, integration, and deployment of end-to-end software systems across frontend applications, backend services, REST APIs, databases, authentication, system integration, and application infrastructure. My experience combines formal software engineering and computer science education with extensive hands-on product development, independent engineering, and technical delivery across the rail sector over 12 cycles.

My engineering work increasingly centres on AI application development, including LLM integration, Retrieval-Augmented Generation (RAG), semantic search, embedding pipelines, prompt and context engineering, agentic architectures, tool calling, and AI-driven workflows. I build the surrounding application and data infrastructure required to make AI systems useful in practice, including document processing, retrieval pipelines, structured outputs, API integrations, evaluation, state management, and controlled execution.

I work across AWS, Microsoft Azure, and Google Cloud, developing cloud-native and distributed systems using serverless and containerised workflows, event-driven architectures, CI/CD pipelines, infrastructure services, security controls, observability, and production-oriented deployment patterns. Alongside cloud engineering, I develop event-driven workflow and integration systems with n8n, combining APIs, webhooks, asynchronous processing, orchestration, reliability controls, monitoring, and custom application interfaces to connect and automate business and technical processes.

Education

- Generative AI**
2017 - 2026 · Stellenbosch University
Intermediate AI training focused on modern AI concepts, generative models, AI application development, and practical applications of emerging AI technologies.
- Full-Stack Software Engineering & Web Development**
2017 - 2022 (Coursework Completed) · University of South Africa
Full-stack software engineering and web development training covering frontend, backend, databases, APIs, software architecture, and application deployment.
- BSc Computer Science**
2017 - 2022 (Coursework Completed) · University of South Africa
Computer Science degree studies covering core programming, software development, computer science, and related technical disciplines.
- National Senior Certificate**
2010 - 2015 · Greenside High School

Certifications

AWS Skill Builder: skillbuilder.aws/user/keenosmith
Microsoft Learn: learn.microsoft.com/en-us/users/keenosmith/
Credly: credly.com/users/keeno-smith

- Manage Kubernetes in Google Cloud**
Sep 2026 · Google Cloud
Intermediate hands-on training in Google Kubernetes Engine (GKE) covering Kubernetes deployments with kubectl, cluster and application management, monitoring and debugging, Prometheus, containerised application deployment, Artifact Registry, Kubernetes manifests, and continuous delivery through practical labs and challenge assessment.
- n8n Workflow Automation & AI**
Sep 2026 · n8n
Applied n8n workflow engineering across API integrations, HTTP requests, OAuth2 and API authentication, secure webhooks, pagination, data transformation, conditional and parallel execution, sub-workflows, custom JavaScript/Python, workflow state, and error handling. Extended into AI-powered workflows using AI Agents, tools and memory, with practical implementation of testing, debugging, modular workflow design, versioning, monitoring, recovery, and production-readiness patterns.
- Microsoft Applied Skills: Microsoft Defender XDR**
August 2026 · Microsoft
Demonstrating practical security operations skills with Microsoft Defender XDR, including incident management, endpoint investigation, threat detection and Advanced Hunting with KQL.
- Microsoft Agent Architect**
Cyber Security & Threat Defense
July 2026 · Microsoft × Fortra
Architect-level training focused on designing scalable AI agent solutions using Microsoft Foundry, including agent architecture, project-based implementation, security labs, simulations, prototyping, and production-oriented approaches to enterprise AI.
- AWS AI / ML Scholars Programme AI Practitioner**
AWS AI Programmer Nanodegree
May 2026 · AWS × Udacity
Project-based AI/ML development covering Python, NumPy, Pandas, neural networks, PyTorch, transformer architectures, generative AI, prompt engineering, and AWS AI/ML services.
- Generative AI for Software Engineers**
AI Course for Developers
Dec 2025 · WeThinkCode_

Applied generative AI development covering LLM integration, prompt engineering, AI-assisted software development, agentic workflows, API integration, and incorporating generative AI capabilities into software applications.

Experience

- Software Engineering & Application Development**
2022 - Present | Project-Based · Independent Development
Designed and delivered full-stack software systems from requirements analysis and solution design through architecture, implementation, integration, testing, deployment, and ongoing maintenance. Held direct ownership of technical decisions across the software lifecycle, with responsibility spanning frontend and backend engineering, API and database architecture, authentication, system integration, AI-enabled functionality, product delivery, and continuous improvement.

Full-Stack Software Engineer
Automated MERN Webpage Application
Designed and developed an end-to-end productivity platform using React, Node.js / Express, and MongoDB, with JWT authentication, interconnected resources, account lifecycle management, and cloud deployment. The project covers application architecture, secure API design, data modelling, state management, and production delivery across the frontend and backend.

Core Responsibilities

- Defined functional requirements and translated them into frontend, backend, API, authentication, and data-layer designs.
- Designed architecture across React, Express REST services, authentication middleware, Mongoose models, and MongoDB persistence, and multiple external data providers. Owned implementation across data integration, asynchronous workflows, persistent state, search, external services, and browser-based functionality.

Core Responsibilities

- Translated functional requirements into frontend architecture, backend services, API contracts, data models, and integration workflows.
- Designed MongoDB persistence using Mongoose for user libraries, resources, and persistent state.
- Implemented client-side state management, asynchronous resource orchestration, validation, error handling, and client-server state reconciliation.
- Maintained source control, dependency management, linting, testing, troubleshooting, and iterative improvements using Git and GitHub.

Full-Stack Software Engineer / AI Engineer
Knowledge Engineering & Generative AI Application
Designed and engineered a full-stack AI-enabled application integrating a React frontend, Node.js / Express backend, MongoDB persistence, document processing, semantic retrieval, local embedding infrastructure, conversational context, and streamed LLM inference. Held direct ownership of the surrounding application architecture and AI data pipeline rather than limiting the implementation to model API integration.

Core Responsibilities

- Implemented local semantic embedding infrastructure using Hugging Face Transformers, ONNX, and all-MiniLM-L6-v2, including lazy model loading, mean pooling, and embedding generation for knowledge chunks and search queries.
- Developed document processing for TXT, Markdown, PDF, and DOCX content using Multer, PDF Parse, and Marmot, including multipart uploads, content extraction, validation, normalisation, chunking, and persistence.
- Integrated server-side LLM inference through the Groq SDK and implemented streamed responses using asynchronous provider streams and Server-Sent Events, with incremental client-side parsing, completion handling, error processing, and request cancellation.
- Designed and maintained MongoDB/Mongoose models and REST API workflows for conversations, messages, memories, knowledge documents, embedded vectors, semantic search, document ingestion, and streamed AI inference.

Full-Stack Software Engineer / Database Application Engineer
Enterprise Database Management
Designed and engineered a full-stack database management system centred on PostgreSQL, SQL, relational modelling, Prisma ORM, and runtime schema management. Owned technical architecture across the browser interface, REST services, database layer, schema management, SQL execution, and metadata synchronisation.

Core Responsibilities

- Designed the PostgreSQL architecture and relational models, including primary and foreign keys, unique constraints, nullability, defaults, identity columns, and relationship integrity.
- Developed Express REST services supporting relational CRUD, dynamic record operations, schema management, relationship operations, database health monitoring, and local SQL execution.
- Developed a dedicated SQL builder and validation layer for DDL generation, including identifier quoting and escaping, data-type allowlisting, type normalisation, and constraint validation.
- Engineered runtime schema operations supporting table creation and modification of columns, constraints, and relationships through PostgreSQL DDL.
- SQL execution through POST /api/query, with PostgreSQL-aware integrity validation, conflict handling, state verification, and repeatable database seeding.

- Technical & Digital Operations Lead | Co-Founder**
Jan 2020 - June 2020 · Firdaus Entertainment
Co-founded and operated an events and entertainment business with responsibility spanning digital product and customer experience, technical and digital operations, business systems, workflow design, project delivery, brand strategy, marketing technology, and operational optimisation.

Responsibilities

- Led the development and ongoing optimisation of the company's digital ecosystem, spanning web presence, digital platforms, business tools, analytics, content systems, and user engagement. A Discovery Agent constructs a resource and
- Translated business objectives and operational requirements into digital solutions, structured workflows, and practical technology implementations.
- Applied systems thinking to analyse operational processes, identify inefficiencies, and design improved workflows and technology-enabled ways of working.
- Researched, evaluated, configured, and integrated digital tools and third-party services to support business operations, communications, marketing, analytics, and process efficiency.
- Managed digital initiatives end-to-end, from requirements definition and solution planning through implementation, testing, troubleshooting, iteration, and user adoption.
- Leveraged data, performance metrics, analytics, and operational feedback to evaluate digital initiatives and drive continuous improvement.
- Worked across web technologies, digital platforms, data and analytics tools, automation, content systems, and integrated services to maintain and evolve the company's digital infrastructure.
- Owned technical and operational problem-solving across the company's digital environment, independently researching solutions, resolving issues, and implementing improvements.
- Managed multiple concurrent digital and operational projects, coordinating dependencies, external providers, resources, timelines, and delivery requirements.

- Contract Specialist | Technical Content & Documentation**
Feb 2021 - Sep 2025 · 3Play Media
Worked extensively with technically complex documentation, training material, and technology-focused content across enterprise software, cloud computing, networking, cybersecurity, software development, and emerging technologies. Responsible for transforming and quality-assuring technical material while maintaining terminology accuracy, contextual integrity, structural consistency, and technical clarity across complex subject matter.

Responsibilities

- Reviewed, transcribed, edited, and proofread technical documentation and training content covering software development, cloud infrastructure, networking, cybersecurity, enterprise technology, and emerging technologies.
- Worked extensively with technical terminology, product specifications, software concepts, infrastructure terminology, security concepts, and technology-specific language across diverse subject areas.
- Analysed source material for inconsistencies, ambiguity, terminology conflicts, structural issues, and technical inaccuracies before producing final content. Combined AST parsing, remediation and verification. The APIs, asynchronous workflows, policy controls, observability, evaluation and human-in-the-loop execution. The architecture uses agentic AI to software engineering and cloud architecture while maintaining deterministic controls, bounded permissions and auditable decision workflows.

Core Responsibilities

- Developed content across enterprise software, cloud computing, networking, cybersecurity, software development, and emerging technologies. Responsible for transforming and quality-assuring technical material while maintaining terminology accuracy, contextual integrity, structural consistency, and technical clarity across complex subject matter.
- Maintained strict quality and consistency standards across technical documentation, including terminology, formatting, structure, instructional clarity, and contextual accuracy.
- Adapted to different technical domains and technology ecosystems, requiring rapid acquisition of domain-specific knowledge and the ability to work effectively with unfamiliar technologies.
- Worked with technical content associated with major enterprise technology organisations and platforms, including Cisco, Oracle, Palo Alto Networks, Unreal Engine, Deloitte, ThousandEyes, Salesforce, NVIDIA, CBT Nuggets, and Apple.
- Operated independently across contract-based assignments, managing quality, accuracy, deadlines, and deliverables while working with highly specialised technical material.

Cloud, DevOps & Platform Engineering

- AWS**
Cloud development and applied AI/ML across AWS services including EC2, S3, Lambda, IAM, VPC, RDS, ECR, ECS, EKS, Kubernetes, SageMaker, and Bedrock. Experience developed through hands-on labs, simulated cloud environments, and project-based learning with AWS Skill Builder.

AWS Skill Builder: skillbuilder.aws/user/keenosmith

- Microsoft**
Cloud and AI development across Azure services including App Service, Functions, Storage, Azure SQL, Cosmos DB, AKS, Azure Container Registry, Entra ID, Azure Monitor, Microsoft Foundry, Azure AI services, and Microsoft Defender XDR. Experience developed through hands-on labs, simulated enterprise environments, and project-based learning with Microsoft Learn.

Microsoft Learn: learn.microsoft.com/en-us/users/keenosmith/

- Google Cloud**
Cloud and AI/ML development across Compute Engine, Cloud Storage, Cloud Run, GKE, Kubernetes, Cloud Functions, Cloud SQL, Firestore, BigQuery, Vertex AI, Gemini, Artifact Registry, Cloud Build, and Google Cloud IAM. Experience developed through hands-on labs, simulated cloud environments, and developer projects, and structured practical learning with Google Skills and Google Developers Program.

Google Skills: skills.google/public_profiles/105079cb-276f-4644-9c48-9c6bf594527
Google Developer Program: g.dev/keenosmith

Projects

Portfolio: keenosmith.vercel.app
GitHub: github.com/keenosmith-del

- Software Engineering**
Full-Stack Development
Frontend Development
Backend Development
API Development
Application Architecture
Component Architecture
State Management

- AI & Agentic Systems**
AI Application Development
Generative AI
Agentic AI
Multi-Agent Systems
LLM Integration
RAG & Knowledge Retrieval
Tool Calling
AI Evaluation

- Data & Knowledge Engineering**
MongoDB & Mongoose
PostgreSQL & SQL
Database Design
Data Modelling
Semantic Embeddings
Vector Retrieval

- Cloud, DevOps & Delivery**
Cloud Deployment
CI/CD
Git & GitHub
Docker
Kubernetes
Authentication & Authorization
Observability
Testing & Evaluation

- Multi-Agent AI Systems - Microsoft Foundry / Azure**
Agentic AI · Multi-Agent Architecture
OpenAI Models · RAG · Tool Calling · OpenAPI · MCP · Python · Azure RBAC · OpenTelemetry
Multi-agent cloud engineering systems developed with Microsoft Foundry and Azure, using specialised agents, tool-augmented reasoning, OpenAPI integrations, structured embeddings, vector retrieval, structured APIs, asynchronous workflows, policy controls, observability, evaluation and human-in-the-loop execution. The architecture uses agentic AI to software engineering and cloud architecture while maintaining deterministic controls, bounded permissions and auditable decision workflows.

Core Responsibilities

- Multi-agent software engineering and DevOps platform separating repository analysis, source-code inspection, application diagnostics and operational investigation into specialised engineering agents. Integrates GitHub repositories, source code, pull requests, documentation, application logs, monitoring telemetry, deployment history and engineering knowledge to analyse software behaviour, investigate failures, correlate evidence and generate structured remediation plans. Agent tool calling, RAG, API integrations and controlled workflow orchestration allow AI agents to operate against real software-development and operational systems rather than functioning as a standalone coding assistant.

GitHub: github.com/keenosmith-del/microsoft-devops-intellect-intelligence

- AI-Powered Cloud Architect & FinOps Optimization Platform**
Multi-agent Azure cloud architecture platform that evaluates infrastructure across cost, performance, reliability, security, governance, operational excellence and sustainability. A Discovery Agent constructs a resource and dependency model using Azure Resource Manager and Resource Graph, while specialised Cost, Reliability, Security and Performance agents independently evaluate the environment before an Architect Agent reconciles competing architectural constraints. Produces structured optimisation findings, projected cost and savings models, architectural impact assessments and controlled remediation plans, with policy validation and human approval separating probabilistic AI recommendations from deterministic decision workflows.

GitHub: github.com/keenosmith-del/microsoft-ai-cloud-architect-finops-optimization

- Cloud Security, DevOps & Agentic AI - AWS**
Cloud Security · Agentic AI · Event-Driven Architecture · DevOps
SRE · Lambda · DynamoDB · CloudWatch · CodeDeploy · IAM · KMS · Docker · EventBridge · BedRock · CloudTrail · S3 · SQS · Fargate · EC3
AWS-native cloud engineering platforms combining event-driven architecture, serverless services, containerised delivery, security automation, agentic AI, workflow orchestration, observability, and controlled automated recovery. The projects demonstrate how AWS-native infrastructure and deterministic engineering controls can be combined with Bedrock and AgenticCore to build secure, stateful and production-oriented systems across cloud security operations and software delivery reliability.

Core Responsibilities

- Event-driven multi-agent security operations architecture for cloud threat detection, investigation, triage and controlled remediation. Uses CloudTrail and EventBridge to detect and route security events into SQS and Lambda before Step Functions coordinates structured incident workflows, parallel investigation, BedrockAgentCore tool calling, risk assessment, human approval, remediation and verification. Combines least-privilege IAM, deterministic security controls, bounded agent permissions, retries, logging and CloudWatch observability, structured incident state, immutable evidence preservation, audit trails and CloudWatch observability.

GitHub: github.com/keenosmith-del/aws-cloud-security-incident-response

- Docker-Distributed Developer Deployment Platform**
Docker-first AWS DevOps and SRE platform automating the software delivery lifecycle while continuously evaluating deployed application health and initiating controlled recovery workflows. Uses GitHub, CodePipeline, CodeBuild, Docker, ECR and ECS/Fargate to build, validate, publish and deploy immutable container artefacts, with CloudWatch telemetry feeds and EventBridge integration. Combines synthetic API transactions, OpenAPI and BedrockAgentCore-assisted root-cause analysis. Combines deterministic reliability guardrails with bounded AI reasoning to support promotion, hold and rollback decisions, alongside retries, dead-letter queues, idempotent operations, observability and controlled automated recovery.

GitHub: github.com/keenosmith-del/aws-distributed-developer-deployment

- Google Cloud**
Cloud-Native Engineering · DevSecOps · Supply Chain Security
Multi-Agent Systems · Gemini · Vertex AI · Firestore · Cloud KMS · BigQuery · Cloud Run · Dataplex · Artifact Registry · Google AIK
Google Cloud-native engineering platforms combining containerised application development, event-driven architecture, specialised AI agents, software supply-chain analysis, enterprise data governance, security controls, and scalable serverless workflows. The projects demonstrate enterprise data integration, cloud-native services to build evidence-driven security and governance workflows across software repositories, application dependencies, enterprise data estates, and cloud infrastructure.

Core Responsibilities

- Software supply-chain security and DevSecOps platform that constructs a machine-readable representation of application dependencies before combining deterministic security analysis with AI/ML-based multi-agent reasoning. Analyses direct and transitive dependencies, package metadata, vulnerabilities, licences, container configuration, build pipelines and application architecture, then correlates findings into structured security assessments and prioritised remediation recommendations. Uses Cloud Build, Artifact Registry and Cloud Run to establish a containerised software delivery path, with Pub/Sub and Cloud Run Jobs supporting asynchronous repository and security analysis workflows.

GitHub: github.com/keenosmith-del/gcp-supply-chain-risk

- Autonomous Data Governance & Compliance Platform**
Enterprise data governance control plane designed to continuously discover, classify and assess heterogeneous data assets across Cloud Storage, BigQuery and Firestore. Combines metadata management, Sensitive Data Protection, access analysis, policy evaluation, risk and confidence scoring, and evidence-based AI reasoning to produce structured governance findings and route approved actions through controlled remediation workflows. Uses specialised AIK agents for discovery, classification, policy and governance analysis while maintaining deterministic controls for sensitive-data detection, policy enforcement, IAM evaluation, auditability and human approval.

GitHub: github.com/keenosmith-del/gcp-data-governance-compliance

- Claude-Powered AI Engineering - Anthropic**
LLM Application Engineering · Cloud Engineering · Semantic Search · Anthropic AI
JSON · AST Parsing · Knowledge Graphs · AI Evaluation · PostgreSQL · Node.js · RAG · OpenAPI
Claude-powered engineering platforms combining long-context reasoning, structured generation, retrieval, semantic analysis, static code analysis, architecture modelling, and decision workflows. The projects use Claude for evidence-grounded reasoning over complex technical and regulatory information, with deterministic processing, structured data pipelines, evaluation, human review, and locally containerised application infrastructure surrounding the model layer.

Core Responsibilities

- Regulatory analysis platform that processes evolving regulatory documents, identifies material changes between versions, extracts obligations, and maps those changes to organisational controls, policies, and operational processes. Combines document ingestion, semantic comparison, local embeddings, vector retrieval, structured Claude outputs, evidence grounding, impact assessment, control mapping, human review, and remediation planning through a custom React and Python/Node.js application.

GitHub: github.com/keenosmith-del/claude-reg-change-impact

- Claude-Powered Architecture Analysis & Decision Engineering Platform**
Software architecture analysis platform that reconstructs system structure from source repositories and infrastructure definitions before using Claude to reason over architecture boundaries, dependencies, coupling, scalability, reliability, and design trade-offs. Combines AST parsing, dependency extraction, architecture graph construction, static analysis, repository and infrastructure inspection, structured architectural assessment, workflow comparison, decision modelling, and automated Architecture Decision Record generation.

GitHub: github.com/keenosmith-del/claude-architecture-analysis

- Workflow Orchestration & Automation - n8n**
Event-Driven Architecture · API Integration · Webhooks · TypeScript · Redis
Docker · OpenTelemetry · GitHub · Express · Prometheus · Grafana
Workflow orchestration and integration platforms developed with n8n as the event-driven automation layer, combining webhooks, API orchestration, asynchronous processing, state management, reliability patterns, observability, containerised infrastructure, and custom local applications. The projects demonstrate enterprise workflow design, heterogeneous system integration, API operations, fault handling, automated remediation, and production-oriented engineering beyond conventional no-code automation.

Core Responsibilities

- Enterprise workflow orchestration platform coordinating complex business processes across heterogeneous systems through event-driven workflows and API integration. Uses n8n for webhook-driven orchestration, conditional routing, sub-workflows, asynchronous execution, retries, compensation patterns, human approval, and error handling, with a custom React and Node.js operations console providing process visibility, execution tracking, approvals, audit trails, and integration health.

GitHub: github.com/keenosmith-del/n8n-opsflow-business-orch

- Webhooks API Orchestration & Integration Reliability Engine**
API operations platform for continuously validating, monitoring, and operating external integrations through webhook-driven workflows and automated reliability controls. Combines synthetic API transactions, OpenAPI and JSON Schema contract validation, API gateway integration, latency and error analysis, retry and backoff strategies, fault injection, integration state management, automated remediation, and operational observability through a custom React and Python/Node.js application.

GitHub: github.com/keenosmith-del/n8n-webhooks-api-orch

- Full-Stack Software Projects**
Full-Stack Engineering · Application Architecture
React · Vite · Node.js · Express · MongoDB · PostgreSQL · Prisma
A portfolio of end-to-end full-stack systems demonstrating frontend architecture, backend service development, REST API design, database architecture, authentication and authorization, state management, asynchronous application flows, component architecture, and integration between client applications, backend services, databases, and external APIs.

Projects span document and relational data models, JWT-based authentication, protected API workflows, Mongoose and Prisma data access, PostgreSQL schema and SQL operations, third-party API integration, external service integration, AI and LLM integration, retrieval workflows, validation, error handling, and application lifecycle management across independently structured frontend and backend systems.

Portfolio: keenosmith.vercel.app

Skills

GitHub: github.com/keenosmith-del

Portfolio: keenosmith.vercel.app

CodeWars: codewars.com/users/keenosmith-del
LeetCode: leetcode.com/u/keenosmith/
FreeCodeCamp: freecodecamp.org/keenosmith

Microsoft Learn: learn.microsoft.com/en-us/users/keenosmith/
AWS Skill Builder: skillbuilder.aws/user/keenosmith/
Google Developer Program: g.dev/keenosmith

Credly: credly.com/users/keenosmith-del

Languages

JavaScript · Python · Java · SQL · HTML · CSS

Frontend & Backend Engineering

React · Vite · Node.js · Express.js · REST APIs · API Design · Authentication & Authorisation · OpenAPI

AI Application Development

Generative AI · LLM Integration · RAG · Embeddings · Vector Search · Prompt & Context Engineering · Agentic AI · Multi-Agent Systems · Tool Calling

Cloud & AI Platforms

Microsoft Azure · Microsoft Foundry · Azure OpenAI · AWS · Amazon Bedrock · Bedrock AgentCore · Amazon SageMaker AI · Google Cloud · Vertex AI · Serverless · Containers · Event-Driven Architecture

DevOps, Automation & SRE

Docker · Git · GitHub · GitHub Actions · CI/CD · Workflow Orchestration · Infrastructure & Deployment Automation · Observability · Incident Response

Data & Security

MongoDB · PostgreSQL · Mongoose · Prisma · Database Design · IAM · RBAC · Least Privilege · Cloud Security · Security Automation